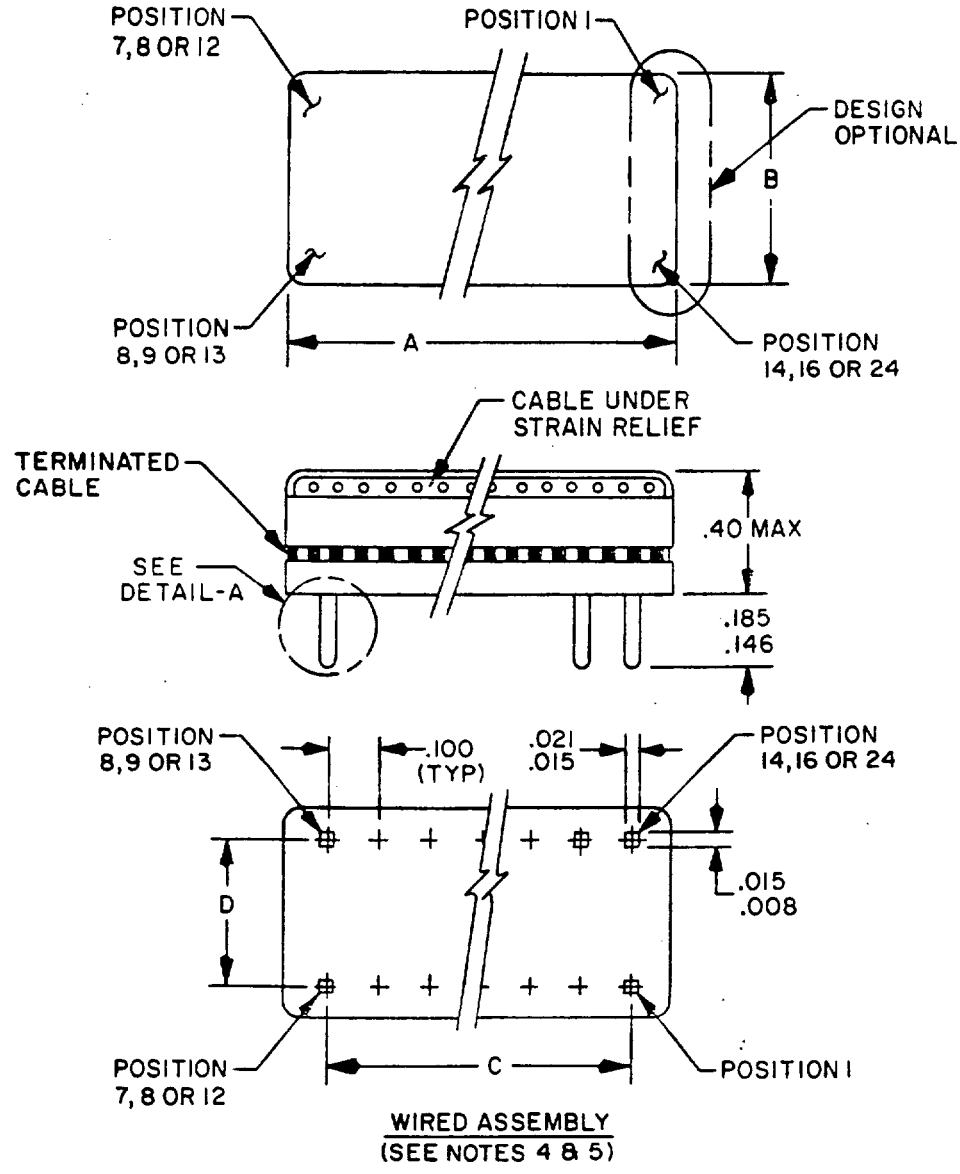
MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL, DUAL-IN-LINE PLUG FOR TERMINATING ROUND CONDUCTORS ON .050 CENTERS, FOR MATING WITH STANDARD DIP SOCKETS, .100 X .300 CONTACT SPACING, .100 X .400 CONTACT SPACING AND .100 X .600 CONTACT SPACING.

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

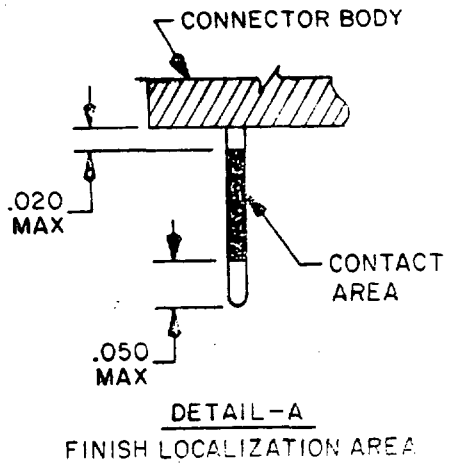
The complete requirements for acquiring the flat cable connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-83503(USAF).



WIRED ASSEMBLY
(SEE NOTES 4 & 5)

FIGURE 1. Dimensions and configuration.

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INCHES	MM	INCHES	MM
.008	.20	.700	17.78
.015	.38	.73	18.5
.020	.51	.78	19.8
.021	.53	.800	20.32
.050	1.27	.88	22.4
.100	2.54	1.000	25.40
.146	3.71	1.100	27.94
.185	4.70	1.18	30.0
.300	7.62	1.28	32.5
.40	10.2	1.300	33.02
.48	12.2	1.48	37.6
.53	13.5	1.900	48.26
.600	15.24	2.08	52.8

Dash numbers and characteristics.

Dash number	No. of connections	Dimensions A (max)	B (max)	C	D
01	14	.78	.48	.600	.300
02	16	.88	.48	.700	.300
03	24	1.28	.73	1.100	.600
04	8	.48	.48	.300	.300
05	18	.98	.48	.800	.300
06	22	1.18	.53	1.000	.400
07	28	1.48	.73	1.300	.600
08	40	2.08	.73	1.900	.600

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals, ±.01 (.3 mm) for two place decimals and ±2° on angles.
4. Strain relief shall be provided. Strain relief design is optional (may be either separate or an integral part of cover).
5. Assembled height including installed cable and (when applicable) the strain relief and excluding leads shall be .40 (10.2 mm) maximum.
6. Tolerances are non-cumulative.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1.

Interface: Round conductor flat cable, .050 centers, to dual-in-line socket .100 x .300 inch, .100 X .400 or .100 X .600 inch.

Temperature range: -55°C to 120°C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Beryllium copper per QQ-C-530 or QQ-C-533 or phosphor bronze (per QQ-B-750) with gold plating in the contact area.

Accessories: May be made of 301 stainless steel or other materials per MIL-C-83503.

Housing: Glass filled nylon per MIL-M-20693 or MIL-M-19887: diallyl phthalate, type SDG-F per MIL-M-14; or glass filled polyester per MIL-M-24519.

Conductor accommodation: 28 AWG stranded .05 spacing between conductor centerlines.

Test wire size: 28 AWG stranded.

Assembly tool: M22520/32.

Dielectric withstanding voltage (at sea level): 500 V rms.

(at altitude): 200 V rms.

Contact resistance: 50 milliohms, maximum.

Humidity: Connector shall be mated with the appropriate series 2 (low profile) dip socket per MIL-S-83734 with long axis of contacts running in a horizontal direction and the cable hanging down vertically.

Polarization: Not required.

Marking:

Contact identification: As a minimum, mark the end contacts in each row with raised numbers or ink stamps on connector cover.

Part number: M83503/6 - (dash number from figure 1).

Intended use: Connector may be directly soldered to a .093 PW board or mated with the appropriate series 2 (low profile) dip socket per MIL-S-83734. These connectors when mated with MIL-S-83734 dip sockets are not to be used in airborne applications.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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